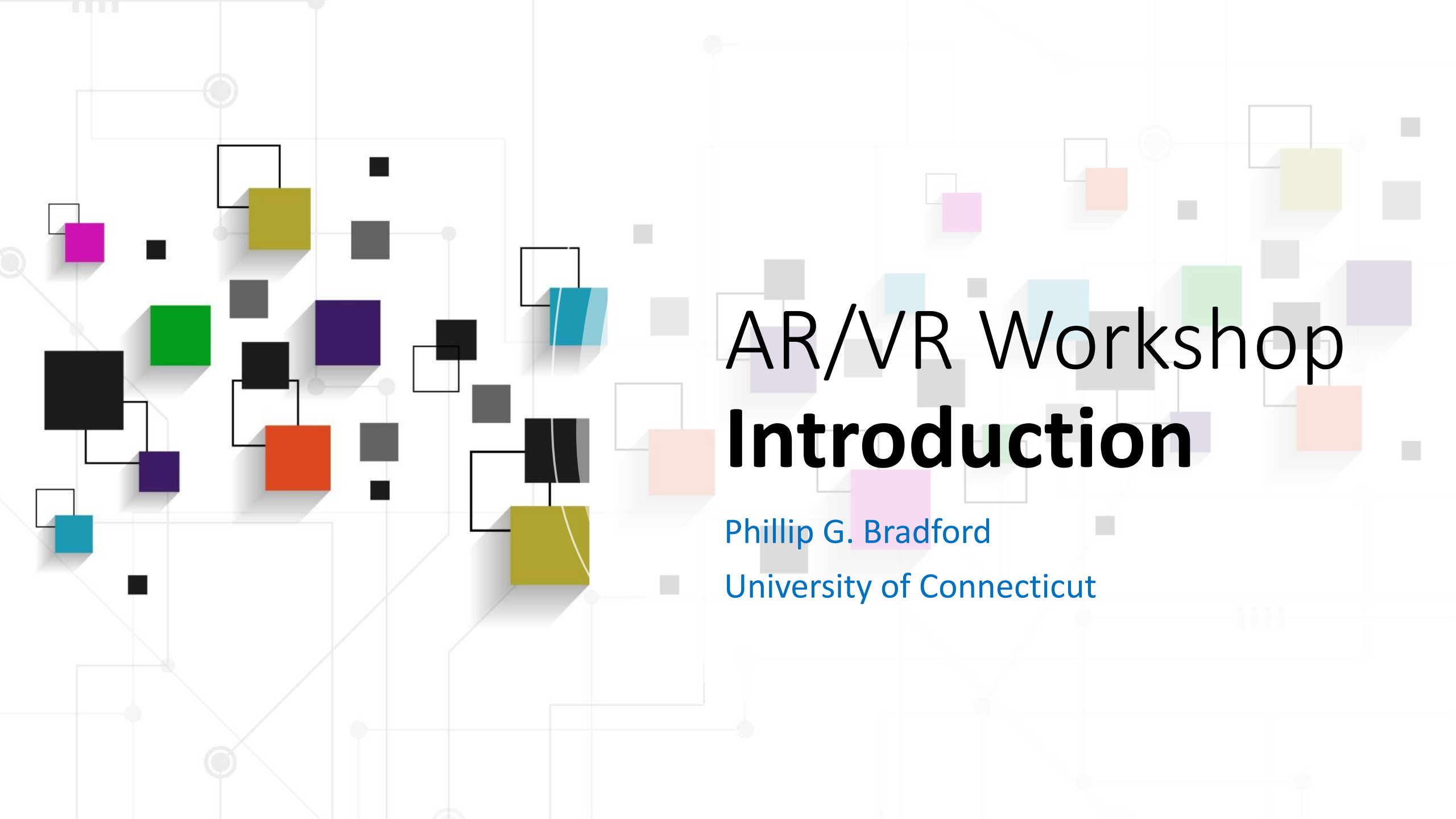


The background of the slide is a light gray grid with various colored squares (black, white, gray, yellow, green, blue, red, purple, orange) scattered across it. Some squares are solid, while others are outlined. Thin black lines connect some of the squares, creating a network-like structure. The overall aesthetic is modern and technical.

AR/VR Workshop Introduction

Phillip G. Bradford

University of Connecticut

Full select demo – how you can do ...



[Basic-Mars-Landing](#)

Module demo highlights: 3D image



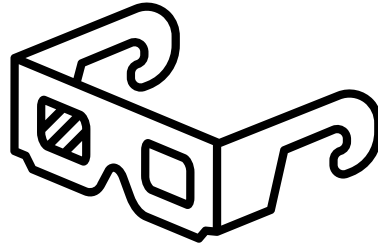
Outline

Overview

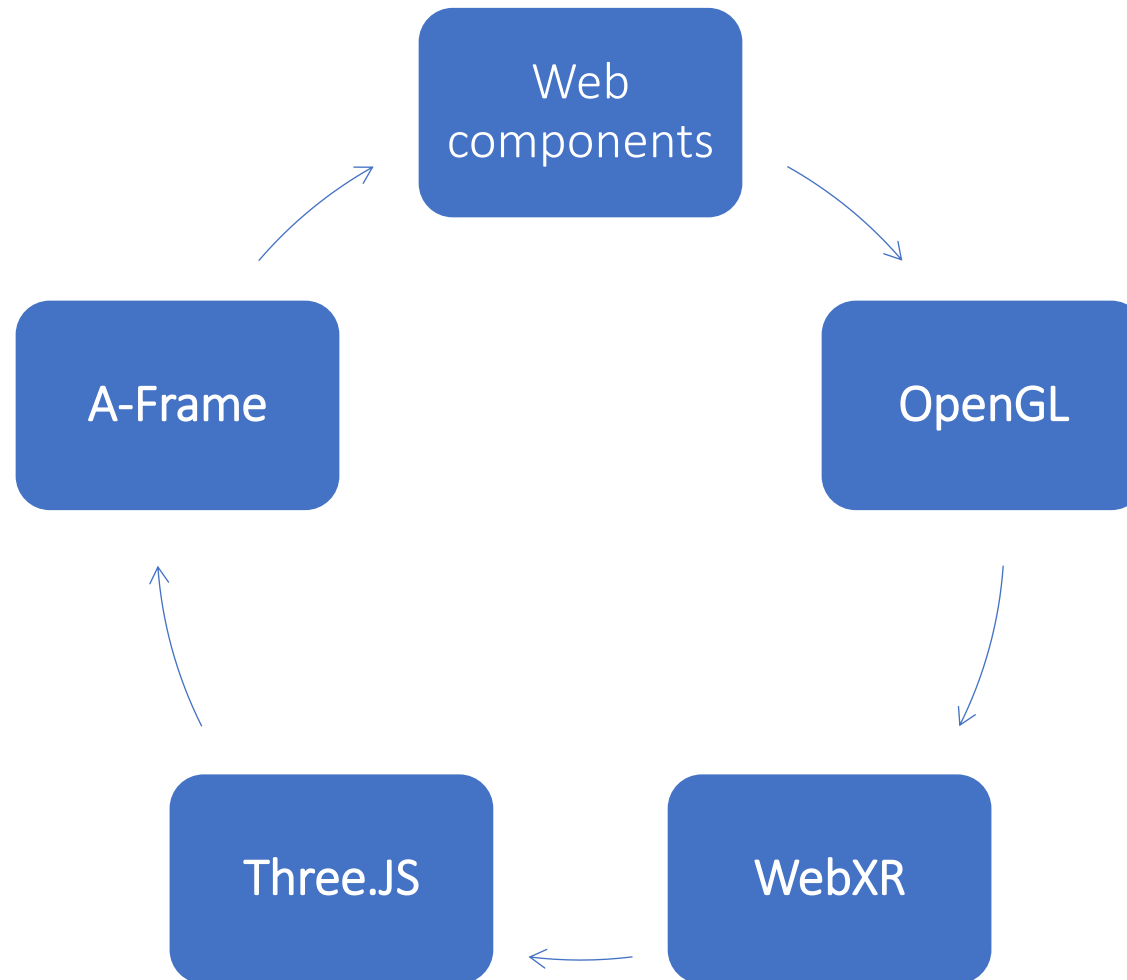
No installations needed

Github pages

Our first VR/AR examples



High level view and learning path



A-frame

3D virtual reality system built by Mozilla

A-frame is a component system built on **Three.js**

Three.js is built on **WebXL** which is built on **WebGL**

A-frame is HTML that uses web-components

Declarative – tell it what you want to see,
it figures out how to do it

No installation VR

Easy

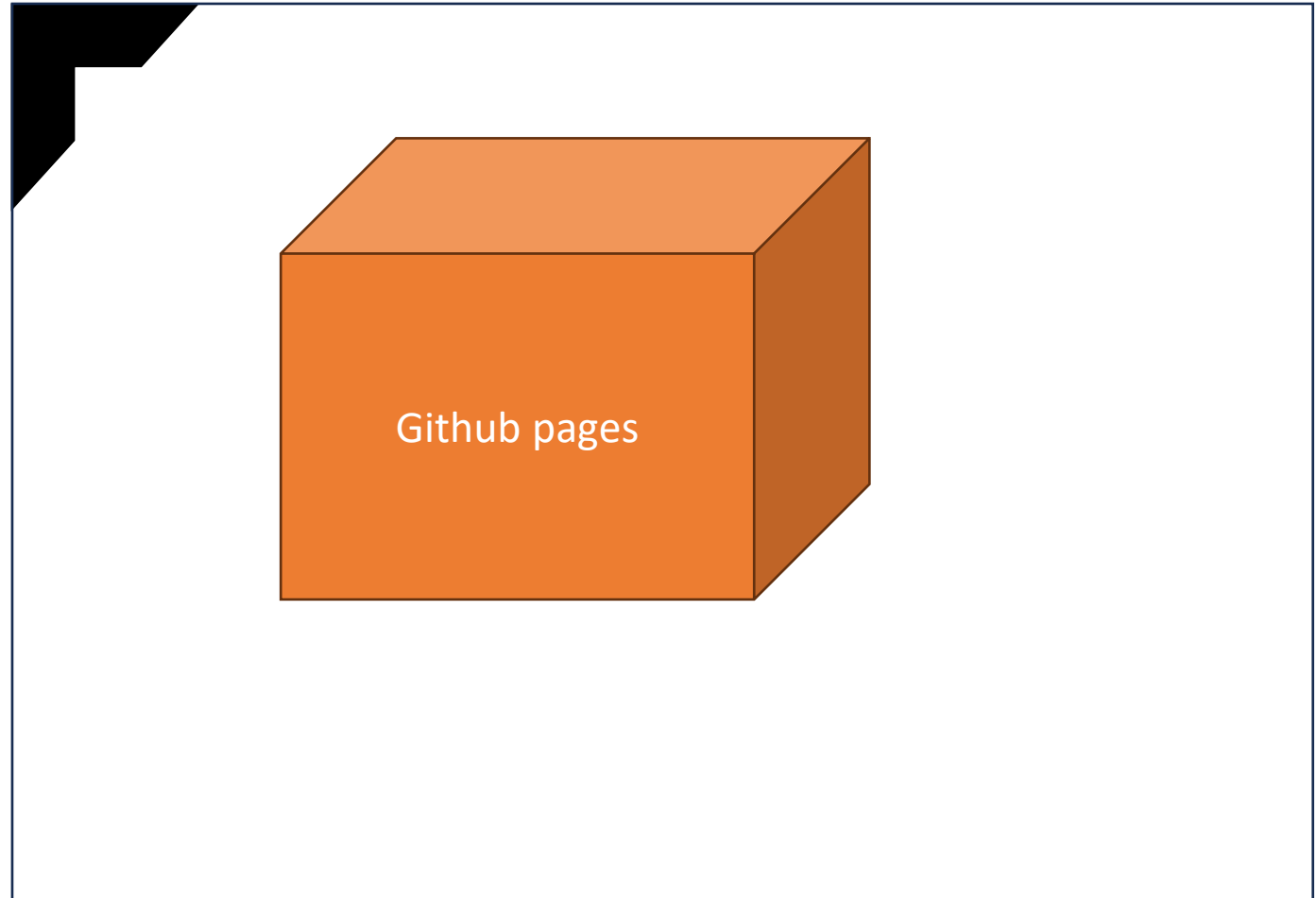
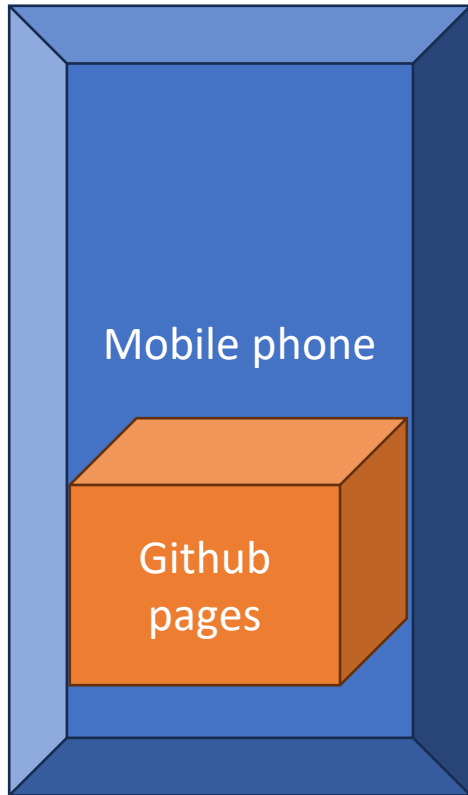
Only Python3 installed

Go to **wonder-phil.github.io** or YOUR_USER.github.io

Clone <https://github.com/wonder-phil/wonder-phil.github.io.git>

Overview: wonder-phil.github.io

Computer



Laptop view

```
git clone https://github.com/wonder-phil/wonder-phil.github.io.git
```

```
cd wonder-phil.github.io
```

```
Terminal> code .
```

```
Terminal> python -m http.server
```

In a browser go to <http://localhost:8000/>

Basic HTML

```
<html>
```

```
  <head>
```

```
    </head>
```

```
    <body>
```

```
      </body>
```

```
    </html>
```

index.html and not-index.html

```
<html>
<head>
  <meta charset="utf-8">
  <title>Hello A-Frame Example</title>
  <script src="https://aframe.io/releases/1.7.1/aframe.min.js"></script>
</head>
<body>
  <a-scene>
    <a-sky color="#0000EE"></a-sky>
    <a-text
      font="kelsonsans"
      color = "#FFFFFF"
      value="HELLO A-Frame!!!"
      width="6"
      position="0 -1 -5"
    ></a-text>
  </a-scene>
</body>
</html>
```

<a-scene>

```
<a-sky color="#0000EE">
</a-sky>
<a-text
  font="kelsonsans"
  color = "#FFFFFF"
  value="HELLO A-Frame!!!"
  width="6"
  position="0 -1 -5"
></a-text>
</a-scene>
```

VR glasses steps

On your phone go to your github pages

wonder-phil.github.io

Use VR glasses to view the project on your phone

simple-sky.html

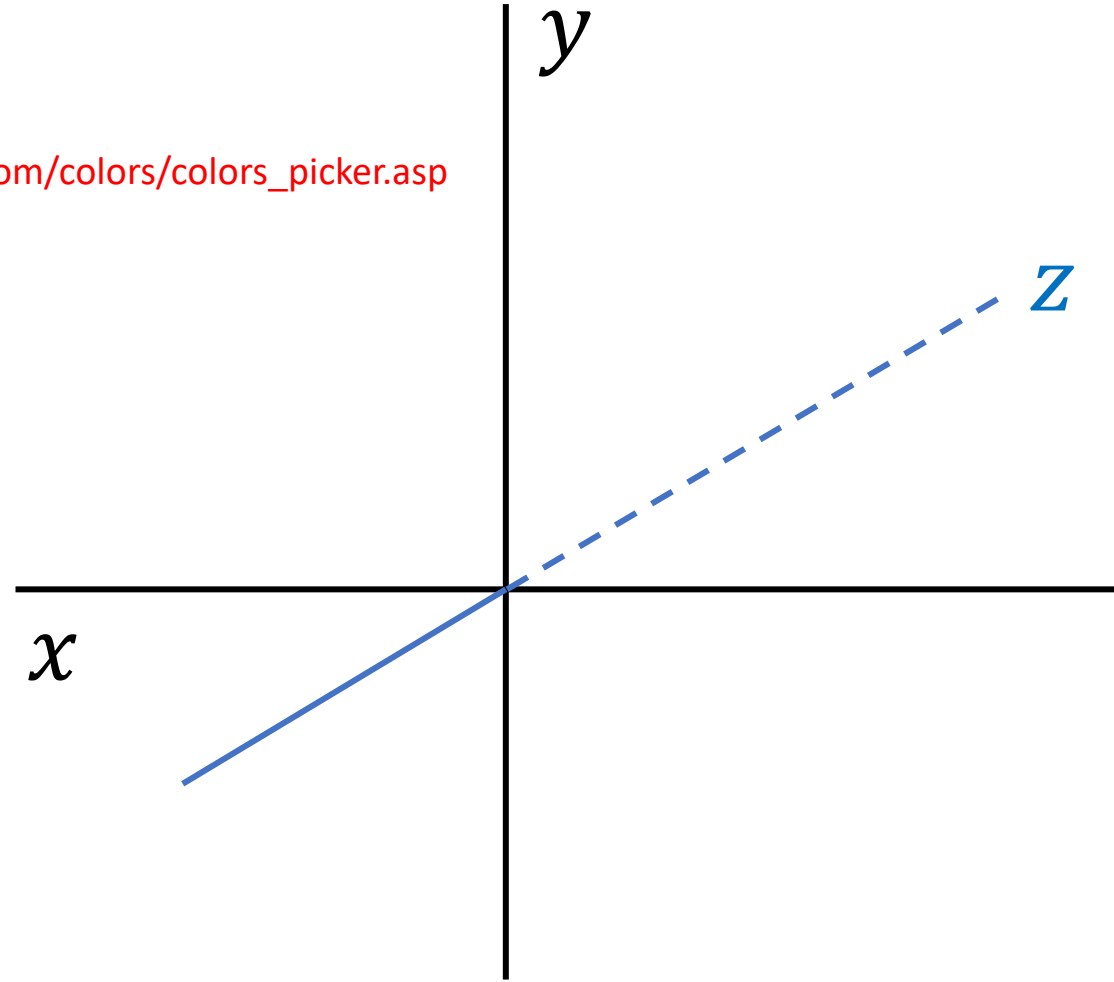
```
<a-scene>
```

```
  <a-sky color="#0105EF">    Select a color at https://www.w3schools.com/colors/colors\_picker.asp  
  </a-sky>
```

```
  <a-text  
    font="kelsonsans"  
    color = "#FFFFFF"  
    value="HELLO A-Frame!!!"  
    width="6"  
    position="0 -1 -5"  Note: x-horizontal, y-vertical, and z-depth*  
  ></a-text>
```

```
</a-scene>
```

*this translates to 1 meter below the origin and 5 meters into the page



simple-sky.html: changing the view

```
<a-scene>  
  <a-sky color="#0105EF">  
  </a-sky>  
  <a-text  
    font="kelsonsans"  
    color = "#FFFFFF"  
    value="HELLO A-Frame!!!"  
    width="6"  
    position="0  -1  -5"  
  ></a-text>  
</a-scene>
```

Replace the current index.html on glitch
with not-index.html
Simple-sky.html

← a-scene now looks like this

Putting it all together

You have now created virtual reality text on a 3D background

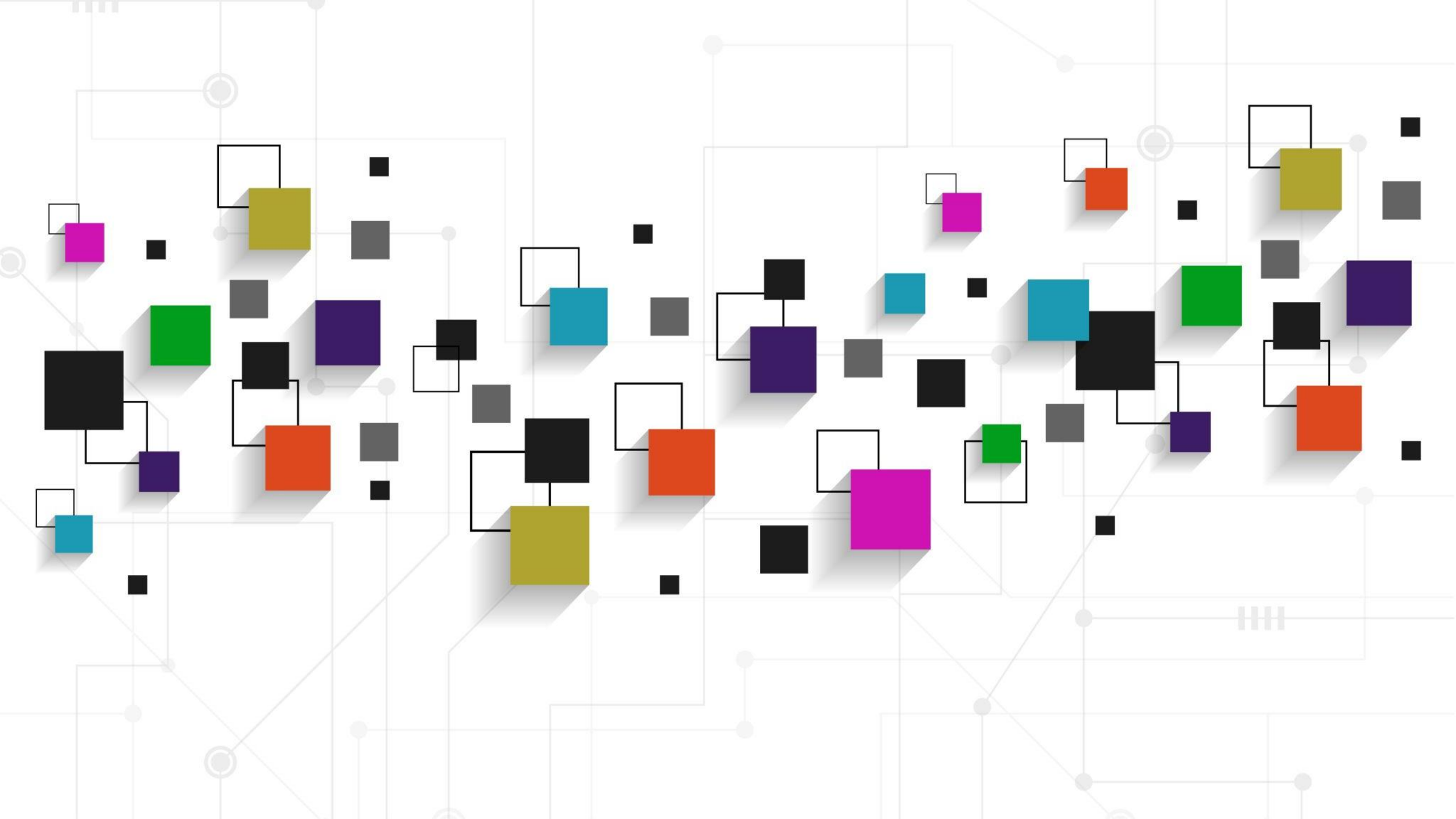
We can also

1. Change background – to the nebula

What if it doesn't work? TROUBLESHOOTING!

2. Move text around in the nebula

What if it doesn't work? TROUBLESHOOTING!



Getting a-frame from aframe.io

```
<head>  
  <script src="https://aframe.io/releases/1.7.1/aframe.min.js">  
  </script>  
</head>
```

Or to run outside of header to help TROUBLE SHOOTING

```
<body>  
  <script src="../../js/aframe-master.min.js"> <!-- OLD a-frame here -->  
                                          <!-- NEW a-frame in head -->  
  </script>
```

TROUBLESHOOTING!

CTRL-ALT-I in browser for a-frame debugging

CMD-Option-I in browser for a-frame debugging

SHIFT-CTRL-J in browser for Chrome/FireFox debugging

CTRL-O in browser to load a file